Cardiology III: Hypertension and Dyslipidemia

All key points for the exam (2025 AHA/ACC Hypertension guideline, 2017 ACC/AHA, 2018 AHA/ACC Cholesterol guideline + 2022 ECDP, ESC) — drug tables included

PART A — HYPERTENSION (HTN)

1. Definition, Measurement and Evaluation

BP category (adults)	Systolic		Diastolic
Normal	< 120	and	< 80
Elevated	120–129	and	< 80
Stage 1 hypertension	**130–139**	or	**80–89**
Stage 2 hypertension	**≥ 140**	or	**≥ 90**
Severe hypertension (no acute target-organ damage — "urgency")	**> 180**	and/or	**> 120**
Hypertensive emergency	> 180	and/or	> 120 **+ acute target-organ damage**

- If SBP and DBP fall in different categories → use the **higher** category • diagnosis based on **average of ≥ 2 readings on ≥ 2 occasions**.
- **Proper measurement:** no caffeine, exercise, or smoking for 30 min • empty bladder • **seated quietly 5 min**, back supported, **feet flat**, legs uncrossed, **arm supported at heart level** • correct cuff size (bladder encircles 80% of arm; small cuff → falsely high) • no talking • measure both arms initially (use higher).
- **Out-of-office monitoring** (home BP or 24-h ambulatory) to confirm diagnosis and titrate: **white-coat HTN** (office ↑, home normal — no treatment, monitor) • **masked HTN** (office normal, home ↑ — treat) • home BP ≥ 130/80 = HTN.
- **Primary (essential) HTN** ~90% • **secondary HTN** ~10% — suspect if resistant, sudden onset, onset < 30 years, severe, hypokalemia, target-organ damage disproportionate.
- **Target-organ damage:** brain (stroke, TIA, dementia), heart (**LVH**, HF, CAD/MI), kidney (**CKD**, albuminuria), eyes (**retinopathy**), **PAD**, aortic aneurysm.
- **Baseline labs:** BMP (K, Na, SCr/eGFR), fasting glucose/A1C, lipid panel, TSH, **urine albumin-to-creatinine ratio**, CBC, Ca, uric acid, ECG.

Secondary causes of hypertension

Cause	Clues / screening
Primary aldosteronism (most common secondary endocrine cause)	Resistant HTN, **hypokalemia**, adrenal incidentaloma, OSA → **aldosterone-to-renin ratio** • 2025 guideline: screen all with resistant HTN and broadly in HTN • treat with **MRA** or adrenalectomy
Renal parenchymal disease (CKD)	↑ SCr, albuminuria
Renovascular (renal artery stenosis)	Abdominal bruit, **↑ SCr > 30% after ACEi/ARB**, flash pulmonary edema, young women (fibromuscular dysplasia)
Obstructive sleep apnea	Snoring, daytime sleepiness, obesity → CPAP
Pheochromocytoma	Episodic HTN, **headache, sweating, palpitations** → plasma free metanephrines • α-blocker (phenoxybenzamine) **before** β-blocker
Cushing's syndrome, thyroid disease, hyperparathyroidism, coarctation of aorta	Specific features
Drug-induced	**NSAIDs**, **decongestants** (pseudoephedrine, phenylephrine), **stimulants** (amphetamines, methylphenidate, caffeine), **oral contraceptives** (estrogen), **corticosteroids**, **cyclosporine/tacrolimus**, **erythropoietin**, **venlafaxine/SNRIs**, bupropion, MAOIs (+ tyramine), **VEGF inhibitors** (bevacizumab, TKIs), **alcohol**, **cocaine**, **licorice**, ginseng, ma huang (ephedra), atypical antipsychotics, testosterone, abrupt clonidine/β-blocker withdrawal

2. Goals and When to Start Medications (2025 AHA/ACC)

- **BP goal < 130/80 mmHg for all adults** (including diabetes, CKD, CAD, HF, stroke, older adults) • encourage **SBP < 120** when tolerated (SPRINT, ESPRIT) • older community-dwelling adults ≥ 65: SBP < 130 • pregnancy and frailty individualized.
- **Risk estimation:** 2025 guideline uses the **PREVENT** equations (10-year CVD risk; replaces Pooled Cohort Equations) for ages 30–79.
- **Elevated BP or stage 1 without high risk → lifestyle**, reassess in 3–6 months.
- **Stage 1 (130–139/80–89) → start medication** if **clinical CVD**, **diabetes**, **CKD**, or **PREVENT 10-year CVD risk ≥ 7.5%** (2017: ASCVD risk ≥ 10%) • otherwise lifestyle × 3–6 months → start medication if BP still ≥ 130/80.
- **Stage 2 (≥ 140/90)** → lifestyle + **medication** — start **2 first-line drugs of different classes**, preferably as a **single-pill combination (SPC)** (when BP > 20/10 above goal) • 2025 favors SPCs for most patients needing ≥ 2 drugs (↑ adherence).
- **Follow-up:** monthly after starting/changing therapy until at goal, then every 3–6 months • check **K and SCr 2–4 weeks** after starting/titrating diuretics, ACEi, ARB, MRA.

3. Lifestyle Modifications

Intervention	Recommendation	Approx. SBP ↓
Weight loss	Ideal body weight • ≥ 5% loss (≈ **1 mmHg per 1 kg**)	**~5 mmHg**
DASH diet	Fruits, vegetables, whole grains, low-fat dairy, ↓ saturated fat/total fat	**~11 mmHg** (largest)
Sodium reduction	Ideally **< 1,500 mg/day**, or **↓ by at least 1,000 mg/day** (2025: < 2,300 mg general target)	5–6 mmHg
Potassium	**3,500–5,000 mg/day** from diet (or **K-enriched salt substitutes**) — unless CKD/hyperkalemia risk	4–5 mmHg
Physical activity	Aerobic **90–150 min/week**, dynamic resistance, isometric (handgrip)	4–8 mmHg
Alcohol	Men ≤ 2, women ≤ 1 drink/day (2025: ideally abstain or minimize)	4 mmHg
Others	Smoking cessation (CV risk), stress management, sleep 7–9 h, treat OSA	—

4. Pharmacotherapy

First-line drug classes

- **Thiazide-type diuretics** (**chlorthalidone preferred** — longer acting, proven outcomes; indapamide; HCTZ), **ACE inhibitors**, **ARBs**, **dihydropyridine CCBs** (amlodipine) — all reduce CV events.
- **Black patients without HF or CKD** → initial **thiazide-type diuretic or CCB** (ALLHAT) — more effective than ACEi/ARB monotherapy (combination SPC often needed).
- **Never combine ACEi + ARB** (or with aliskiren) → ↑ hyperkalemia, AKI, hypotension, no benefit (ONTARGET, VA NEPHRON-D).
- **β-blockers are NOT first-line** for uncomplicated HTN (less stroke protection, esp. atenolol) — use for **compelling indications** (HFrEF, post-MI/CAD, AF rate control, migraine, essential tremor, hyperthyroidism, pregnancy — labetalol).
- **Alpha-1 blockers**, central α2 agonists, direct vasodilators, loop diuretics = **add-on/later-line** agents.
- Preferred SPC combinations: **ACEi/ARB + CCB** (ACCOMPLISH — best), **ACEi/ARB + thiazide**, CCB + thiazide.

Compelling indications

Condition	Preferred agents	Avoid / caution
HFrEF	**ARNI/ACEi/ARB + evidence-based β-blocker + MRA + SGLT2i** + diuretics • H-ISDN (Black patients)	**Non-DHP CCBs**, clonidine (↑ mortality), α-blockers (moxonidine), minoxidil
HFpEF	Diuretics (volume), SGLT2i, MRA, ARB/ARNI, then other first-line agents	—
Stable CAD / post-MI	**β-blocker + ACEi/ARB** (+ DHP CCB, nitrates for angina)	Non-DHP CCB with β-blocker; IR nifedipine
Diabetes	Any first-line agent • **ACEi/ARB if albuminuria** (UACR ≥ 30)	—
CKD with albuminuria (UACR ≥ 300, or ≥ 30)	**ACEi or ARB** (max tolerated) + SGLT2i ± finerenone	ACEi + ARB combination • K and SCr monitoring
Stroke/TIA (secondary prevention)	**Thiazide + ACEi/ARB** (PROGRESS) • goal < 130/80	—
Pregnancy	**Labetalol**, **nifedipine ER**, **methyldopa** • treat if ≥ 140/90 (CHAP) • low-dose aspirin 81 mg from 12 weeks (preeclampsia prevention in high risk)	**ACEi, ARBs, ARNI, aliskiren, MRA — contraindicated** (fetotoxic) • atenolol (growth restriction) • avoid diuretics initiation
Women of child-bearing potential	CCB, labetalol • counsel if on ACEi/ARB (stop when planning pregnancy)	ACEi/ARB without contraception
AF (rate control)	β-blocker or non-DHP CCB	—
BPH	**α1-blocker** (doxazosin, terazosin) — add-on only, not monotherapy (ALLHAT ↑ HF)	—
Migraine / essential tremor	Propranolol (β-blocker)	—
Osteoporosis	**Thiazides** (↓ urinary Ca, ↑ bone density)	Loop diuretics
Gout / hyperuricemia	**Losartan** (uricosuric), CCBs	**Thiazides and loop diuretics** (↑ uric acid)
Asthma/COPD	Any first-line agent • cardioselective β-blocker if needed	**Nonselective β-blockers** (propranolol, nadolol)
Hyperkalemia risk	Thiazide, CCB	ACEi, ARB, MRA, K-sparing diuretics
Raynaud's	DHP CCB	β-blockers
Elderly / orthostatic hypotension	Start low; CCB, thiazide, ACEi/ARB	**α-blockers**, central agents (Beers), clonidine

Resistant hypertension

- **Definition:** BP above goal despite **3 drugs of different classes** (including a **diuretic**, ideally long-acting thiazide-type) at **maximally tolerated doses**, or BP controlled but requiring **≥ 4 drugs**.
- **First exclude pseudoresistance: non-adherence** (most common), improper measurement, **white-coat effect** (confirm with ABPM/home), suboptimal doses, **interfering drugs** (NSAIDs, decongestants, OCs, etc.), high sodium/alcohol intake.
- **Screen for secondary causes** — especially **primary aldosteronism** and **OSA**, CKD, renal artery stenosis.
- **Management:** maximize diuretic (switch **HCTZ → chlorthalidone or indapamide**; loop if eGFR < 30 — though chlorthalidone works in CKD stage 4, CLICK) → **add spironolactone 25–50 mg** (best 4th-line agent — PATHWAY-2) or eplerenone → then amiloride, β-blocker (or combined α/β), hydralazine, clonidine, minoxidil • **aprocitentan** (dual endothelin antagonist, Tryvio — teratogenic, edema/anemia) • **renal denervation** in selected patients • lifestyle/sodium restriction.

5. Antihypertensive Drug Classes — Key Points

Class	Examples (daily dose)	Key ADRs / notes
Thiazide-type diuretics	**Chlorthalidone 12.5–25 mg** • **HCTZ 25–50 mg** • **indapamide 1.25–2.5 mg** • metolazone 2.5–5 mg	**Hypokalemia**, **hyponatremia** (elderly women), **hypomagnesemia**, **hypercalcemia**, **hyperuricemia/gout**, **hyperglycemia**, hyperlipidemia, photosensitivity, sulfa allergy (low cross-reactivity), ED • **less effective eGFR < 30** (except chlorthalidone/metolazone) • take in the morning • ↓ lithium clearance (↑ lithium toxicity)
ACE inhibitors	**Lisinopril 10–40**, enalapril 5–40, ramipril 2.5–20, benazepril 10–40, captopril (TID), quinapril, perindopril	**Dry cough** (10–20%, bradykinin — switch to ARB), **angioedema** (rare, life-threatening, ↑ in Black patients — **never rechallenge**), **hyperkalemia**, **↑ SCr** (accept ≤ 30%), hypotension, **CI pregnancy (boxed)**, bilateral RAS, history of angioedema, with sacubitril (36 h) • ↑ lithium • NSAIDs ↓ effect/↑ AKI ("triple whammy" with diuretic)
ARBs	**Losartan 50–100**, valsartan 80–320, irbesartan 150–300, olmesartan 20–40, candesartan 8–32, telmisartan 20–80	Same as ACEi **without cough** (angioedema rare — can use cautiously after ACEi angioedema) • CI pregnancy • **losartan ↓ uric acid** • olmesartan: sprue-like enteropathy
DHP CCBs	**Amlodipine 2.5–10**, nifedipine ER 30–90, felodipine 2.5–10	**Peripheral (ankle) edema** (dose-related; ↓ by adding ACEi/ARB — not diuretics), headache, flushing, **reflex tachycardia**, **gingival hyperplasia** • avoid IR nifedipine • felodipine/nifedipine: CYP3A4 substrates (grapefruit)
Non-DHP CCBs	**Diltiazem ER 120–360**, **verapamil ER 120–480**	**Bradycardia**, AV block, **constipation (verapamil)**, edema • **CI in HFrEF** • avoid with β-blockers • **CYP3A4 inhibitors** (↑ simvastatin/lovastatin, DOACs, cyclosporine) • P-gp (↑ digoxin)
β-blockers	β1-selective: **metoprolol**, **atenolol**, bisoprolol • nonselective: propranolol, nadolol • α/β: **carvedilol**, **labetalol** • vasodilating: **nebivolol** (NO)	Bradycardia, fatigue, ↓ exercise tolerance, **bronchospasm** (nonselective), **masked hypoglycemia** (tachycardia/tremor — not sweating), sexual dysfunction, depression, cold extremities, ↑ TG/↓ HDL • **abrupt withdrawal → rebound HTN/angina/MI** (taper 1–2 weeks) • atenolol renally cleared
Mineralocorticoid receptor antagonists	**Spironolactone 25–50**, eplerenone 50–100 (BID)	**Hyperkalemia**, **gynecomastia**/breast tenderness, menstrual irregularities (spironolactone) • resistant HTN, primary aldosteronism, HFrEF • eplerenone CYP3A4
K-sparing diuretics	Amiloride 5–10, triamterene 50–100	Hyperkalemia • triamterene: kidney stones • usually combined with thiazide
Loop diuretics	Furosemide 20–80 BID, torsemide 5–10, bumetanide 0.5–2 BID	HTN with **eGFR < 30** or HF • hypokalemia, ototoxicity, gout
α1-blockers	**Doxazosin** 1–8, **prazosin** 2–20 (BID–TID), **terazosin** 1–20	**First-dose orthostatic hypotension/syncope → start at bedtime**, low dose • dizziness • not monotherapy (↑ HF — ALLHAT) • BPH benefit • Beers (avoid as antihypertensive in elderly)
Central α2 agonists	**Clonidine 0.1–0.8 mg BID** (patch weekly 0.1–0.3 mg/day), **methyldopa 250–1,000 mg BID**, guanfacine	Sedation, **dry mouth**, dizziness, bradycardia, depression • **clonidine: rebound HTN with abrupt withdrawal** (taper; worse with β-blocker → stop β-blocker first) • **methyldopa: pregnancy**; **hemolytic anemia** (+ Coombs), hepatitis, lupus-like • Beers

Class	Examples (daily dose)	Key ADRs / notes
Direct vasodilators	**Hydralazine 25–100 mg BID–TID**, **minoxidil 5–100 mg**	**Reflex tachycardia** and **Na/water retention** → combine with **β-blocker + diuretic** • hydralazine: **drug-induced lupus** (slow acetylators, > 200 mg/day), headache • minoxidil: **hirsutism**, **pericardial effusion**, severe edema (resistant HTN)
Direct renin inhibitor	Aliskiren 150–300	Hyperkalemia, diarrhea, angioedema • **CI pregnancy** • **CI with ACEi/ARB in diabetes**
Endothelin antagonist	Aprocitentan 12.5 mg	Edema, anemia, hepatotoxicity • **embryo-fetal toxicity** • resistant HTN

Counseling and monitoring

- Take diuretics in the **morning** (nocturia) • rise slowly (orthostasis) • don't stop β-blockers or clonidine abruptly • avoid NSAIDs and decongestants • home BP monitoring (validated upper-arm device) • limit salt, avoid K salt substitutes with ACEi/ARB/MRA in CKD • report swelling of lips/tongue (angioedema) immediately.
- Monitor **BP, HR, K, Na, SCr, glucose, uric acid, lipids** depending on drug.

6. Hypertensive Crises

- **Severe hypertension without acute target-organ damage** ("urgency") — BP > 180/120, asymptomatic or minor symptoms (headache): **no need for IV drugs or ICU • reinstitute/intensify oral therapy** (often non-adherence), reduce gradually over **24–48 h to days**, follow-up within 1 week • **avoid rapid lowering** (ischemia) — **never use IR (sublingual) nifedipine**.
- **Hypertensive emergency** — BP > 180/120 **+ acute target-organ damage**: hypertensive encephalopathy, stroke/ICH, ACS, **acute pulmonary edema/HF**, **aortic dissection**, **AKI**, **eclampsia/severe preeclampsia**, retinal hemorrhage/papilledema, microangiopathic hemolytic anemia → **ICU + IV titratable agents**.
- **BP lowering targets (general):** ↓ SBP by **no more than 25% within the first hour** → then to **160/100–110** over the next **2–6 h** → then to normal over **24–48 h**.
- **Exceptions (compelling conditions): aortic dissection → SBP < 120 within 20 minutes** (and HR < 60) — **β-blocker first** (esmolol, labetalol) then vasodilator (avoid reflex tachycardia) • **severe preeclampsia/eclampsia** or **pheochromocytoma crisis → SBP < 140 in the first hour • acute ischemic stroke** → < 185/110 before thrombolysis, < 180/105 for 24 h after; if no lysis, treat only if > 220/120 (↓ 15%) • **acute ICH** → SBP target ~140 (avoid < 130).

IV agent	Dose	Key points
Nicardipine	**5 mg/h** → ↑ 2.5 mg/h every 5–15 min (max 15 mg/h)	DHP CCB • most conditions (stroke, AKI) • reflex tachycardia, headache • avoid in acute HF
Clevidipine	1–2 mg/h → double every 90 s (max 32 mg/h)	Ultra-short DHP CCB • **lipid emulsion → CI soy/egg allergy**, defective lipid metabolism • max 1,000 mL/24 h
Labetalol	0.3–1 mg/kg (max 20 mg) IV bolus q10 min, or 0.4–1 mg/kg/h infusion (max 3 mg/kg/h)	α1 + β blocker • **pregnancy**, aortic dissection, stroke • avoid in asthma, HF, bradycardia, heart block
Esmolol	500–1,000 mcg/kg load → 50–300 mcg/kg/min	β1-selective, ultra-short (RBC esterases) • **aortic dissection**, perioperative • bradycardia
Nitroprusside	0.3–0.5 mcg/kg/min → max 10 mcg/kg/min	Arterial + venous dilator, immediate onset • **cyanide toxicity** (hepatic impairment, > 2 mcg/kg/min, prolonged — give **sodium thiosulfate/hydroxocobalamin**) • **thiocyanate toxicity** (renal failure) • **protect from light** • ↑ ICP (avoid in stroke) • arterial line
Nitroglycerin	5 mcg/min → max 200 mcg/min	Venodilator • **ACS and acute pulmonary edema** (adjunct) • headache, tachyphylaxis

IV agent	Dose	Key points
Hydralazine	10–20 mg IV q4–6h	Unpredictable response, reflex tachycardia • **pregnancy** (preeclampsia) • avoid in ACS/dissection
Enalaprilat	1.25–5 mg IV q6h	Variable response, slow onset • **CI pregnancy**, bilateral RAS • HF
Fenoldopam	0.1–0.3 mcg/kg/min	Dopamine-1 agonist • ↑ renal blood flow • ↑ intraocular pressure (avoid glaucoma)
Phentolamine	5 mg IV bolus q10 min	α-blocker • **catecholamine excess**: pheochromocytoma, cocaine/amphetamine, MAOI–tyramine crisis, clonidine withdrawal

- **Agent selection: aortic dissection** → esmolol/labetalol (+ nicardipine/nitroprusside after β-blockade) • **ACS** → nitroglycerin, esmolol/labetalol (avoid hydralazine) • **acute pulmonary edema** → nitroglycerin/nitroprusside + loop diuretic (avoid β-blockers) • **stroke/ICH** → nicardipine, clevidipine, labetalol • **preeclampsia/eclampsia** → **labetalol, hydralazine, nifedipine IR (oral, allowed in pregnancy)** + **magnesium sulfate** (seizures) • **pheochromocytoma/cocaine** → phentolamine, nicardipine, benzodiazepines (cocaine) — avoid unopposed β-blockade • **AKI** → nicardipine, clevidipine, fenoldopam.

Hypertension in special populations

- **Pregnancy:** chronic HTN (before 20 weeks) — treat ≥ 140/90 (goal < 140/90) with **labetalol, nifedipine ER, methyldopa** • preeclampsia (≥ 20 weeks + proteinuria/organ dysfunction) • **low-dose aspirin 81 mg** from 12 weeks in high-risk • severe (≥ 160/110) → IV labetalol, IV hydralazine, oral IR nifedipine within 30–60 min.
- **Elderly:** start low, go slow • watch orthostatic hypotension, falls, hyponatremia (thiazides) • avoid central α-agonists, α-blockers (Beers) • goal SBP < 130 if tolerated.
- **Children/adolescents:** percentile-based (≥ 95th = HTN; ≥ 130/80 in ≥ 13 years) • lifestyle first; ACEi/ARB, CCB, thiazide (not in pregnancy-capable without counseling).
- **CKD:** ACEi/ARB if albuminuria • SGLT2i • loop or chlorthalidone in advanced CKD • accept SCr ↑ ≤ 30% after ACEi/ARB.
- **Black patients:** thiazide or CCB initially (without CKD/HF) • often need ≥ 2 drugs • higher ACEi angioedema risk.

7. Hypertension Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Chlorthalidone	**12.5–25 mg daily**	↓ K, ↓ Na, ↓ Mg, ↑ Ca, ↑ uric acid, ↑ glucose	Preferred thiazide • works in CKD 4 • ↑ lithium • morning
Hydrochlorothiazide	**25–50 mg daily**	Same, photosensitivity	Less effective eGFR < 30 • ↑ skin cancer (non-melanoma) signal
Indapamide	1.25–2.5 mg daily	Same (less metabolic)	Stroke outcomes (PROGRESS)
Lisinopril	**10–40 mg daily**	**Cough**, **angioedema**, ↑ K, ↑ SCr	**CI pregnancy** • K/SCr 2–4 weeks • ↑ lithium
Enalapril / ramipril	5–40 mg / 2.5–20 mg	Same	Ramipril (HOPE)
Losartan	**50–100 mg daily**	↑ K, ↑ SCr, dizziness	**↓ uric acid** • CI pregnancy
Valsartan / olmesartan	80–320 mg / 20–40 mg	Same • olmesartan sprue-like enteropathy	CI pregnancy
Amlodipine	**2.5–10 mg daily**	**Edema**, headache, flushing, gingival hyperplasia	Long half-life • simvastatin ≤ 20 mg
Nifedipine ER	30–90 mg daily	Edema, headache, reflex tachycardia	Pregnancy option • **no IR** for HTN (except obstetric emergency)
Diltiazem / verapamil ER	120–360 / 120–480 mg	Bradycardia, AV block, constipation (verapamil)	**CI HFrEF** • 3A4 inhibitors • ↑ digoxin

Drug	Dose	Key ADRs	Key notes / interactions
Metoprolol succinate	25–400 mg daily	Bradycardia, fatigue	Compelling indications • taper • CYP2D6
Carvedilol	6.25–25 mg BID	Orthostasis	With food • HFrEF
Labetalol	PO 100–400 mg BID (max 2,400/day) • IV 20 mg → 40–80 mg	Orthostasis, hepatotoxicity, bronchospasm	**Pregnancy**
Nebivolol	5–40 mg daily	Headache, fatigue	β1-selective + NO vasodilation
Spironolactone	**25–50 mg daily**	**Hyperkalemia**, **gynecomastia**	**Resistant HTN 4th line**, primary aldosteronism
Eplerenone	50 mg daily–BID	Hyperkalemia	No gynecomastia • CYP3A4
Amiloride	5–10 mg	Hyperkalemia	Resistant HTN • lithium-induced DI
Doxazosin / terazosin	1–8 mg / 1–20 mg at bedtime	**First-dose syncope**, orthostasis	BPH • not monotherapy • Beers
Clonidine	0.1–0.8 mg BID • patch 0.1–0.3 mg/day weekly	Sedation, dry mouth, bradycardia	**Rebound HTN** — taper • Beers
Methyldopa	250–1,000 mg BID–TID	Sedation, **hemolytic anemia**, hepatitis	**Pregnancy**
Hydralazine	PO 25–100 mg BID–TID • IV 10–20 mg	**Lupus**, reflex tachycardia, edema	+ β-blocker + diuretic • pregnancy (IV)
Minoxidil	5–100 mg/day	**Hirsutism**, edema, **pericardial effusion**	Resistant HTN + β-blocker + loop
Aliskiren	150–300 mg	Hyperkalemia, diarrhea	**CI with ACEi/ARB in diabetes**, pregnancy
Nicardipine IV	**5 → 15 mg/h**	Reflex tachycardia, headache	Emergencies (stroke, AKI)
Clevidipine IV	1–2 → 32 mg/h	Hypotension	**CI soy/egg allergy**
Esmolol IV	500 mcg/kg → 50–300 mcg/kg/min	Bradycardia, hypotension	**Aortic dissection**
Nitroprusside IV	0.3 → 10 mcg/kg/min	**Cyanide/thiocyanate**	Protect from light • ↑ ICP

PART B — DYSLIPIDEMIA

8. Lipid Basics and Evaluation

- **Lipoproteins: LDL** (atherogenic, main target), **VLDL** (TG-rich), chylomicrons (dietary TG), **HDL** (reverse cholesterol transport — raising HDL with drugs does **not** ↓ events), **Lp(a)** (genetic, atherogenic, not affected by lifestyle), **apoB** (number of atherogenic particles).
- **Fasting or non-fasting** lipid panel acceptable (fasting if TG ≥ 400) • **Friedewald: LDL = TC − HDL − TG/5** (invalid if TG > 400 → direct LDL or Martin-Hopkins/Sampson equations) • **non-HDL-C = TC − HDL** (secondary target, especially if TG ↑).
- **Desirable values:** LDL < 100 (optimal), TG < 150, HDL > 40 (men)/> 50 (women), total cholesterol < 200 mg/dL • **LDL ≥ 190** = severe (suspect familial hypercholesterolemia) • **TG ≥ 500** (esp. ≥ 1,000) = **pancreatitis risk**.
- **Measure Lp(a) once** in adults (≥ 50 mg/dL or ≥ 125 nmol/L = risk enhancer).
- **Screen** adults ≥ 20 years every 4–6 years (earlier/more often with risk factors); children 9–11 and 17–21 years (universal), FH family screening from age 2.

Secondary causes of dyslipidemia

Cause	LDL ↑	TG ↑
Diet	Saturated/trans fats, weight gain, anorexia	Weight gain, very low-fat diets, refined carbs, **alcohol**
Drugs	**Diuretics**, **cyclosporine**, **glucocorticoids**, **amiodarone**, anabolic steroids, progestins, isotretinoin, SGLT2i (small)	**Oral estrogens**, **glucocorticoids**, **bile acid sequestrants**, **protease inhibitors**, **retinoids** (isotretinoin), **atypical antipsychotics** (olanzapine, clozapine), **β-blockers** (non-vasodilating), **thiazides**, tamoxifen, raloxifene, sirolimus, L-asparaginase, cyclophosphamide
Diseases	**Hypothyroidism**, **nephrotic syndrome**, **obstructive liver disease/cholestasis**, pregnancy	**Diabetes** (poorly controlled), **CKD**, nephrotic syndrome, **hypothyroidism**, obesity/metabolic syndrome, pregnancy, lipodystrophy, HIV

9. Statin Treatment Decisions (2018 AHA/ACC + updates)

Four statin benefit groups

Group	Recommendation
1. Clinical ASCVD (ACS, MI, stable angina, revascularization, stroke/TIA, PAD)	**High-intensity statin** (age ≤ 75) → goal **≥ 50% LDL reduction** • > 75: moderate- or high-intensity • **very high risk** (multiple major ASCVD events or 1 event + multiple high-risk conditions) → add **ezetimibe** if **LDL ≥ 70** (2022 ECDP threshold **≥ 55**) → then **PCSK9 inhibitor**
2. Severe primary hypercholesterolemia — LDL ≥ 190 mg/dL (age 20–75)	**High-intensity statin** without risk calculation → add **ezetimibe** if LDL ≥ 100 (< 50% reduction) → PCSK9 inhibitor (heterozygous FH)
3. Diabetes mellitus, age 40–75, LDL ≥ 70	**At least moderate-intensity statin** • **high-intensity** if multiple risk factors or age 50–75 (ADA: high-intensity with ≥ 1 ASCVD risk factor)
4. Primary prevention, age 40–75, LDL 70–189, no diabetes	Based on **10-year ASCVD risk** (Pooled Cohort Equations; PREVENT newer): • **< 5% (low)** → lifestyle • **5–7.4% (borderline)** → moderate-intensity statin if **risk enhancers** present • **7.5–19.9% (intermediate)** → **moderate-intensity statin** (↓ LDL 30–49%); use risk enhancers/CAC to decide • **≥ 20% (high)** → **high-intensity statin** (↓ LDL ≥ 50%)

- **Risk-enhancing factors: family history of premature ASCVD** (men < 55, women < 65), **LDL ≥ 160** persistently, **CKD**, **metabolic syndrome**, **preeclampsia**/premature menopause (< 40), **chronic inflammatory disorders** (RA, psoriasis, lupus, HIV), **South Asian ancestry**, persistent **TG ≥ 175**, **hs-CRP ≥ 2**, **Lp(a) ≥ 50 mg/dL (≥ 125 nmol/L)**, **apoB ≥ 130**, **ABI < 0.9**.
- **Coronary artery calcium (CAC) score** (intermediate/borderline risk, uncertain): **CAC = 0** → reasonable to **withhold statin** (unless diabetes, smoker, family history of premature ASCVD) and reassess in 5–10 years • **1–99** → favors statin (esp. ≥ 55 years) • **≥ 100 or ≥ 75th percentile** → statin indicated.
- **Age 20–39:** lifestyle; statin if LDL ≥ 160 with family history of premature ASCVD • **age > 75 primary prevention**: clinical judgment (CAC helpful).
- **Clinician–patient risk discussion** before starting statin in primary prevention.
- **LDL targets (ESC 2019/2025; widely used):** very high risk **< 55 mg/dL** (and ≥ 50% ↓; < 40 if recurrent event within 2 years) • high risk **< 70** • moderate risk < 100 • low risk < 116.

Statin intensity

High-intensity (↓ LDL ≥ 50%)	Moderate-intensity (↓ LDL 30–49%)	Low-intensity (↓ LDL < 30%)
Atorvastatin 40–80 mg **Rosuvastatin 20–40 mg**	**Atorvastatin 10–20 mg** **Rosuvastatin 5–10 mg** **Simvastatin 20–40 mg** **Pravastatin 40–80 mg** **Lovastatin 40–80 mg** Fluvastatin XL 80 mg / 40 mg BID **Pitavastatin 1–4 mg**	Simvastatin 10 mg Pravastatin 10–20 mg Lovastatin 20 mg Fluvastatin 20–40 mg

- **Simvastatin 80 mg not recommended** (myopathy) unless already tolerated for ≥ 12 months.
- **Monitoring:** baseline lipids, **ALT** (no routine LFT monitoring afterward unless symptoms), CK only if symptoms/high risk, A1C if diabetes risk • **lipid panel 4–12 weeks after starting or changing dose** → every 3–12 months to assess adherence and response.
- **Administration: simvastatin, lovastatin, fluvastatin IR, pravastatin** (short half-life) → **evening/bedtime** (cholesterol synthesis peaks at night) • **lovastatin IR with evening meal** (food ↑ absorption) • **atorvastatin, rosuvastatin, pitavastatin** (long half-life) → **any time**.

Statin adverse effects

- **Statin-associated muscle symptoms (SAMS): myalgia** (most common, ~5–10%; normal CK) → **myopathy/myositis** (CK ↑) → **rhabdomyolysis** (CK > 10× ULN + renal injury, dark urine) • management: check CK, TSH, vitamin D; **hold statin** until symptoms resolve → **rechallenge** with lower dose, different statin (rosuvastatin/pravastatin/pitavastatin), or **alternate-day dosing** • CoQ10 not proven.
- **Risk factors for myopathy: high dose**, **age > 80**, female, low body weight, **hypothyroidism**, **CKD**, liver disease, **DDIs** (CYP3A4 inhibitors, gemfibrozil), Asian ancestry (rosuvastatin), alcohol, strenuous exercise, SLCO1B1 variants (simvastatin).
- **Hepatotoxicity:** transient ALT ↑ (dose-related, rarely significant) • stop if ALT > 3× ULN with symptoms/jaundice • **contraindicated in decompensated cirrhosis/acute liver failure** (compensated liver disease/MASLD OK).
- **New-onset diabetes** (small ↑ risk, esp. high-intensity) — benefit outweighs risk • rare cognitive complaints (not proven), immune-mediated necrotizing myopathy (anti-HMGCR).
- **Pregnancy and breastfeeding: stop statins** (discontinue when pregnancy planned/detected) — FDA removed strongest contraindication in 2021 but still generally avoid (may continue in very high-risk, e.g., HoFH) • bile acid sequestrants are safe in pregnancy.

Statin drug interactions: simvastatin, lovastatin (most), atorvastatin = **CYP3A4** substrates → **CI with strong 3A4 inhibitors** (clarithromycin, erythromycin, itraconazole, ketoconazole, posaconazole, voriconazole, **HIV protease inhibitors**, cobicistat, nefazodone) and **gemfibrozil**, **cyclosporine**, danazol • **simvastatin max 10 mg** with verapamil, diltiazem, dronedarone • **simvastatin max 20 mg** with **amiodarone**, **amlodipine**, **ranolazine**, lomitapide • lovastatin max 20 mg with diltiazem/verapamil/danazol, max 40 mg with amiodarone • avoid **large amounts of grapefruit juice** (> 1 quart/day) • **pravastatin, rosuvastatin, pitavastatin** = minimal CYP metabolism (fewer DDIs; preferred with PIs) • **rosuvastatin**: start **5 mg in Asian patients**, max 10 mg if **CrCl < 30** (not on dialysis), max 5 mg with cyclosporine • fluvastatin = CYP2C9 • **all statins + gemfibrozil → avoid** (use fenofibrate) • statins slightly ↑ INR with warfarin (rosuvastatin, simvastatin, fluvastatin) • ↑ digoxin (atorvastatin) • colchicine + statin → myopathy.

10. Non-Statin Lipid-Lowering Therapy

Drug	Mechanism / effect	Key points
Ezetimibe (Zetia) 10 mg daily	Inhibits intestinal cholesterol absorption (**NPC1L1**) • ↓ LDL **15–25%**	**First add-on to statin** (IMPROVE-IT ↓ events) • well tolerated (diarrhea, myalgia, ↑ LFTs with statin) • any time, with/without food • cyclosporine ↑ levels • separate from bile acid sequestrants (2 h before/4 h after)

Drug	Mechanism / effect	Key points
PCSK9 monoclonal antibodies: **evolocumab** (Repatha) 140 mg SC q2 weeks or 420 mg monthly • **alirocumab** (Praluent) 75–150 mg q2 weeks or 300 mg monthly	Bind PCSK9 → ↑ LDL receptor recycling • ↓ LDL **50–60%**	ASCVD or FH not at goal on max statin + ezetimibe (FOURIER, ODYSSEY ↓ events) • ADRs: **injection-site reactions**, nasopharyngitis, flu-like • refrigerate (room temp up to 30 days) • expensive
Inclisiran (Leqvio)	**siRNA** → ↓ hepatic PCSK9 synthesis • ↓ LDL ~50%	**284 mg SC at day 0, 3 months, then every 6 months** (given by healthcare provider) • injection-site reactions • outcomes trials ongoing
Bempedoic acid (Nexletol) 180 mg daily (± ezetimibe — Nexlizet)	**ATP-citrate lyase (ACL) inhibitor** (upstream of HMG-CoA reductase) • prodrug activated in **liver, not muscle** → **no muscle symptoms** • ↓ LDL ~18–25%	**Statin-intolerant** patients (CLEAR Outcomes ↓ MACE) • ADRs: **hyperuricemia/gout**, **tendon rupture**, ↑ SCr, anemia, cholelithiasis • ↑ **simvastatin (max 20 mg)** and **pravastatin (max 40 mg)** levels
Bile acid sequestrants: **cholestyramine** 4–24 g/day, **colestipol** 5–30 g (granules) / 2–16 g (tabs), **colesevelam** 3.75 g/day	Bind bile acids in gut → ↑ hepatic LDL receptors • ↓ LDL 15–30% • **↑ TG**	**CI if TG > 400–500**, bowel obstruction, (cholestyramine: hypertriglyceridemia) • ADRs: **constipation**, bloating, gas • **bind other drugs** → take other drugs **1 h before or 4–6 h after** (levothyroxine, warfarin, digoxin, OCs, fat-soluble vitamins A, D, E, K) • colesevelam fewer interactions, also ↓ A1C • **safe in pregnancy** (not absorbed)
Fibrates: **fenofibrate** 48–145 mg (product-specific), **gemfibrozil 600 mg BID** 30 min before breakfast and dinner	PPAR-α agonists → ↑ lipoprotein lipase • **↓ TG 30–50%**, ↑ HDL	**Severe hypertriglyceridemia (≥ 500)** to prevent pancreatitis • no CV benefit added to statins • ADRs: **gallstones**, GI, myopathy (**gemfibrozil + statin → avoid**; fenofibrate preferred with statin), **↑ SCr** (fenofibrate; renal adjust, avoid severe CKD), ↑ LFTs • **↑ warfarin effect** • gemfibrozil: CYP2C8 inhibitor (CI with repaglinide, selexipag)
Niacin (nicotinic acid) ER 500–2,000 mg at bedtime	↓ VLDL production • ↓ TG, ↓ LDL, **↑ HDL (most)**	**No benefit added to statin** (AIM-HIGH, HPS2-THRIVE — ↑ harm) → rarely used • ADRs: **flushing/pruritus** (prostaglandin — take **aspirin 325 mg 30 min before**, take with low-fat snack, avoid hot drinks/alcohol, ER form), **hyperglycemia**, **hyperuricemia/gout**, **hepatotoxicity** (SR forms), GI, myopathy with statins
Omega-3 fatty acids: **icosapent ethyl** (Vascepa, pure EPA) **2 g BID with food** • omega-3 acid ethyl esters (Lovaza, EPA + DHA) 4 g/day	↓ hepatic TG synthesis • ↓ TG 20–45%	**Icosapent ethyl**: ↓ MACE in patients with **ASCVD or diabetes + risk factors** on statin with **TG 150–499** (REDUCE-IT) • ADRs: **AF**, **bleeding**, arthralgia, fishy taste/burping • EPA + DHA products may **↑ LDL** • OTC supplements not equivalent
Lomitapide, **evinacumab**	MTP inhibitor / ANGPTL3 antibody	**Homozygous FH** only • lomitapide: hepatotoxicity (REMS), GI • LDL apheresis

Hypertriglyceridemia management

- **TG 150–499 (moderate): lifestyle** (weight loss, ↓ refined carbs/sugar, ↓ alcohol, exercise), treat secondary causes (diabetes, hypothyroidism, drugs) • statin according to ASCVD risk (persistent TG ≥ 175 = risk enhancer) • **add icosapent ethyl** if ASCVD/diabetes + risk factors with TG 150–499 on statin.
- **TG ≥ 500, especially ≥ 1,000 (severe): pancreatitis prevention is the priority → very low-fat diet** (< 10–15% calories if ≥ 1,000), **alcohol abstinence**, glycemic control, stop offending drugs → **fibrate first-line** (fenofibrate) ± **omega-3 (4 g/day)** → then statin for ASCVD risk • TG-induced acute pancreatitis → insulin infusion, plasmapheresis.
- **Bile acid sequestrants worsen TG** — avoid if TG > 300–400.

Special populations

- **Familial hypercholesterolemia** (LDL ≥ 190, tendon xanthomas, corneal arcus < 45, premature CAD) → high-intensity statin + ezetimibe + PCSK9i • screen family.
- **Children/adolescents with FH:** statin from **age 8–10** if LDL ≥ 190 (or ≥ 160 with risk factors) after lifestyle.
- **Older adults > 75:** continue statin if tolerated; secondary prevention moderate- or high-intensity • primary prevention individualized (CAC).
- **CKD (not on dialysis):** moderate-intensity statin or statin + ezetimibe (SHARP) • **dialysis:** don't initiate statin (no benefit — 4D, AURORA), continue if already on it.
- **HIV:** statin for primary prevention age 40–75 (**pitavastatin** — REPRIEVE ↓ MACE) • prefer pitavastatin/pravastatin/rosuvastatin (PI interactions).
- **Pregnancy/breastfeeding:** stop statins, ezetimibe, PCSK9i • **bile acid sequestrants** are acceptable • women of child-bearing age on statins → contraception.
- **Diabetes:** moderate (40–75) or high-intensity (risk factors/50–75) statin • **stroke**: high-intensity (SPARCL) with LDL < 70.

Lifestyle (all patients)

- Heart-healthy diet: vegetables, fruits, whole grains, legumes, nuts, fish, lean protein • **saturated fat < 5–6% of calories** (reduce), avoid **trans fats**, limit sweets/sugar-sweetened beverages, red/processed meats • **Mediterranean or DASH** • soluble fiber, plant stanols/sterols (2 g/day ↓ LDL ~10%) • **aerobic exercise 150 min/week** • weight loss • smoking cessation • limit alcohol (↑ TG).

11. Dyslipidemia Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Atorvastatin	**High 40–80** / moderate 10–20 mg (any time)	Myalgia, ↑ ALT, ↑ glucose	CYP3A4 • ↑ digoxin • no renal adjust
Rosuvastatin	**High 20–40** / moderate 5–10 mg (any time)	Myalgia, proteinuria	**Asians start 5 mg** • CrCl < 30 max 10 • cyclosporine max 5 • minimal CYP
Simvastatin	Moderate 20–40 mg (evening) • **no 80 mg**	Myopathy (highest)	**CYP3A4** • max 10 mg with verapamil/diltiazem/dronedarone • max 20 mg with amiodarone/amlodipine/ranolazine • CI strong 3A4 inhibitors, gemfibrozil, cyclosporine
Lovastatin	20–80 mg with evening meal	Myopathy	CYP3A4 • max 20 mg with diltiazem/verapamil
Pravastatin	Moderate 40–80 mg (bedtime)	Fewer muscle effects	No CYP • PI/transplant patients
Pitavastatin	1–4 mg (moderate 2–4)	Myalgia	Minimal CYP (UGT) • HIV (REPRIEVE) • CI cyclosporine
Fluvastatin	20–80 mg (XL 80)	Myalgia	CYP2C9 (fluconazole ↑)
Ezetimibe	**10 mg daily**	Diarrhea, myalgia	First add-on • separate from BAS
Evolocumab	**140 mg SC q2 wk or 420 mg monthly**	Injection site, nasopharyngitis	ASCVD/FH not at goal
Alirocumab	75–150 mg SC q2 wk or 300 mg monthly	Injection site	Same
Inclisiran	**284 mg SC day 0, 3 mo, then q6 mo**	Injection site, arthralgia	Administered by HCP
Bempedoic acid	**180 mg daily**	**Gout/hyperuricemia**, **tendon rupture**	Statin-intolerant • simvastatin ≤ 20, pravastatin ≤ 40
Cholestyramine	4–24 g/day (powder in water)	**Constipation**, ↑ TG, ↓ vitamins	**Separate drugs 1 h before/4–6 h after** • CI TG > 400 • pregnancy OK

Drug	Dose	Key ADRs	Key notes / interactions
Colesevelam	3.75 g/day (6 tabs once or 3 BID with meals)	Constipation, ↑ TG	Fewer DDIs (separate levothyroxine, OCs 4 h) • ↓ A1C
Fenofibrate	48–145 mg daily (product-specific)	Gallstones, ↑ SCr, myopathy (less), ↑ LFTs	TG ≥ 500 • OK with statins • renal adjust • ↑ INR
Gemfibrozil	**600 mg BID** 30 min before meals	Gallstones, myopathy	**Avoid with statins** • CYP2C8 inhibitor (CI repaglinide) • ↑ INR
Niacin ER	500 → 2,000 mg at bedtime	**Flushing**, hyperglycemia, gout, hepatotoxicity	Aspirin 325 mg 30 min before • not recommended routinely
Icosapent ethyl	**2 g BID with food**	**AF**, bleeding, arthralgia	TG 150–499 on statin with ASCVD/DM (REDUCE-IT)
Omega-3 acid ethyl esters	4 g daily	Fishy taste, ↑ LDL	TG ≥ 500

Exam Traps and Frequently Tested Points

1. **BP categories:** normal < 120/80 • elevated 120–129/< 80 • **stage 1 130–139 or 80–89** • **stage 2 ≥ 140 or ≥ 90** • use the higher category.
2. **Measure correctly:** seated 5 min, back supported, feet flat, arm at heart level, correct cuff, average ≥ 2 readings on ≥ 2 occasions • confirm with home/ambulatory BP.
3. **White-coat HTN** (office ↑, home normal) → monitor • **masked HTN** (office normal, home ↑) → treat.
4. **Goal < 130/80 for all adults** (SBP < 120 encouraged if tolerated).
5. **Stage 1:** drugs if CVD, diabetes, CKD, or 10-year risk ≥ 7.5% (PREVENT; 10% ASCVD in 2017) • otherwise lifestyle 3–6 months • **stage 2: 2 drugs** (single-pill combination).
6. **Lifestyle:** DASH (~11 mmHg, largest), weight loss (1 mmHg/kg), Na < 1,500 or ↓ 1,000 mg, K 3,500–5,000 mg, exercise 90–150 min/week, alcohol ≤ 1–2 drinks.
7. **First-line: thiazide (chlorthalidone), ACEi, ARB, DHP CCB • Black patients (no CKD/HF): thiazide or CCB** • β-blockers only for compelling indications.
8. **Never ACEi + ARB (+ aliskiren)** • best SPC: ACEi/ARB + CCB.
9. **Pregnancy:** labetalol, nifedipine ER, methyldopa • **CI ACEi/ARB/ARNI/aliskiren** • aspirin 81 mg from 12 weeks for preeclampsia prevention.
10. **CKD with albuminuria → ACEi/ARB** (accept SCr ↑ ≤ 30%) • **gout → losartan**, avoid thiazides • **osteoporosis → thiazide • BPH → α-blocker (not monotherapy) • asthma → avoid nonselective β-blockers**.
11. **Thiazides:** ↓ K, ↓ Na, ↓ Mg, **↑ Ca, ↑ uric acid, ↑ glucose, ↑ lipids** • ineffective eGFR < 30 (except chlorthalidone/metolazone) • ↑ lithium.
12. **ACEi:** cough (bradykinin → switch to ARB), angioedema (never rechallenge), hyperK, ↑ SCr, CI pregnancy and bilateral RAS • check K/SCr in 2–4 weeks.
13. **DHP CCB:** ankle edema (↓ by adding ACEi/ARB), gingival hyperplasia, reflex tachycardia • **non-DHP:** bradycardia, constipation (verapamil), CI HFrEF, CYP3A4 inhibitors.
14. **β-blockers:** mask hypoglycemia (not sweating), bronchospasm (nonselective), **taper — no abrupt stop • clonidine rebound HTN** (taper; stop β-blocker first).
15. **α1-blockers:** first-dose syncope → bedtime • **hydralazine** lupus + reflex tachycardia (add β-blocker + diuretic) • **minoxidil** hirsutism, pericardial effusion • **methyldopa** hemolytic anemia (pregnancy).
16. **Resistant HTN:** ≥ 3 drugs incl. diuretic (or ≥ 4 controlled) • rule out non-adherence, white coat, drugs (NSAIDs, decongestants, OCs) • screen **primary aldosteronism**, OSA • **switch to chlorthalidone → add spironolactone**.
17. **Drug-induced HTN:** NSAIDs, decongestants, stimulants, OCs, steroids, cyclosporine/tacrolimus, EPO, venlafaxine, VEGF inhibitors, licorice, alcohol, cocaine.
18. **Severe HTN without organ damage (urgency):** oral therapy, lower over 24–48 h, **no IR nifedipine • emergency:** IV agents, ↓ ≤ 25% in 1st hour → 160/100–110 over 2–6 h → normal over 24–48 h.
19. **Aortic dissection: SBP < 120 within 20 min, β-blocker first (esmolol/labetalol)** • preeclampsia/pheochromocytoma: SBP < 140 in first hour • stroke: < 185/110 before tPA.

20. **Nitroprusside:** cyanide (hepatic) / thiocyanate (renal) toxicity, protect from light • **clevidipine:** CI soy/egg allergy • **phentolamine:** pheochromocytoma, cocaine, MAOI crisis.
21. **LDL = TC − HDL − TG/5** (invalid TG > 400) • non-HDL = TC − HDL • measure Lp(a) once • TG ≥ 500 → pancreatitis risk.
22. **Four statin groups:** ASCVD → high-intensity • **LDL ≥ 190 → high-intensity** • diabetes 40–75 → moderate (high if risk factors) • primary prevention by 10-year risk (7.5–19.9% moderate, ≥ 20% high).
23. **Risk enhancers:** family history, LDL ≥ 160, CKD, metabolic syndrome, preeclampsia/early menopause, RA/psoriasis/HIV, South Asian, TG ≥ 175, hs-CRP ≥ 2, Lp(a) ≥ 50, apoB ≥ 130, ABI < 0.9.
24. **CAC = 0 → may withhold statin** (except diabetes, smokers, family history) • CAC ≥ 100 → statin.
25. **High-intensity = atorvastatin 40–80, rosuvastatin 20–40** • moderate: atorva 10–20, rosuva 5–10, simva 20–40, prava 40–80, lova 40–80, pitava 1–4.
26. **Simvastatin/lovastatin in the evening** (lovastatin with food) • atorva/rosuva any time • lipid panel 4–12 weeks after start • no routine LFT/CK monitoring.
27. **SAMS:** hold → rechallenge lower dose/different statin/alternate-day • risk: high dose, elderly, hypothyroidism, CKD, DDIs • rhabdo CK > 10× ULN.
28. **Statin DDIs:** CYP3A4 (simva, lova, atorva) — avoid strong inhibitors, gemfibrozil, cyclosporine • **simvastatin ≤ 10 (verapamil, diltiazem), ≤ 20 (amiodarone, amlodipine, ranolazine)** • rosuvastatin 5 mg in Asians.
29. **Add-on order:** statin → **ezetimibe** → **PCSK9 inhibitor** (or inclisiran) • **bempedoic acid** for statin intolerance (gout, tendon rupture).
30. **Bile acid sequestrants:** ↑ TG (CI > 400), constipation, bind drugs (1 h before / 4–6 h after) • pregnancy-safe • colesevelam ↓ A1C.
31. **Fibrates for TG ≥ 500 • gemfibrozil + statin = avoid** (fenofibrate OK) • gallstones, ↑ SCr • niacin: flushing (aspirin 30 min before), hyperglycemia, gout — no added CV benefit.
32. **Icosapent ethyl 2 g BID** → ↓ MACE (TG 150–499 + ASCVD/DM on statin) • AF and bleeding risk.
33. **Statins in pregnancy → stop** • dialysis → don't start • HIV → pitavastatin • FH children from 8–10 years.

Note: recommendations follow the 2025 AHA/ACC hypertension guideline (with 2017 ACC/AHA background), the 2018 AHA/ACC cholesterol guideline with 2022 ECDP updates, and ESC lipid targets. If your course slides or reference use slightly different cut-offs (e.g., 2017 ASCVD ≥ 10% threshold), **use your course numbers for the exam**.